GEOMETRICALLY CONNECTED FIBRES OF A PROPER MORPHISM TOWARDS A NORMAL INTEGRAL BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8304-8344, immediately following the statement at 8286-8302.

Editorial note. Same-name pair note: this is the spaces (more general) side of a label that also exists in chapter more-morphisms; the schemes side is not counted (see dedup.scope).

Editorial note. Three cross-chapter pointer families (Decent Spaces, Cohomology of Spaces, Spaces over Fields) are recorded in assembly.external_references; all same-chapter pointers are line-indexed stubs into the pinned source.

Allowed standard prerequisites (editorial).

- Stein factorization for algebraic spaces. Theorem theorem-stein-factorization-general of this chapter, cited at proof line 8304. Provides the factorization X -> Y' -> Y with Y' the relative spectrum of f_*O_X; the whole proof is the identification Y' = Y.
- Etale-local nature of the normalization and of Y'. Morphisms of Spaces, Lemma spaces-morphisms-lemma-properties-normalization, cited at proof lines 8310-8312. Formation of Y' commutes with etale base change, which lets the problem be moved to an affine integral etale cover V -> Y.
- Covering a normal integral algebraic space by affines. Spaces over Fields, Lemma spaces-over-fields-lemma-normal-integral-cover-by-affines, cited at proof lines 8308-8310. Reduces the isomorphism Y' = Y to checking Y' x_Y V -> V over affine integral schemes V.
- Finite birational morphisms towards a normal scheme are isomorphisms. Morphisms, Lemma morphisms-lemma-finite-birational-over-normal, cited at proof lines 8326-8328. The criterion applied to Y' -> Y once both are normal integral schemes with the same function field.
- Decentness of X over a decent separated base. Decent Spaces, Lemmas decent-spaces-lemma-properties-trivial-implications and decent-spaces-lemma-property-over-property, cited at proof lines 8332-8335. Transfers decency from Y to X so that the generic-point bookkeeping of assumptions (5) and (6) applies to |X|.
- Flat base change for cohomology. Cohomology of Spaces, Lemma spaces-cohomology-lemma-flat-base-change-cohomology, cited at proof lines 8319-8321. Shows condition (6) (H^0 of the generic fibre equals kappa(xi)) is preserved by the etale base change V -> Y.

Source reference. Original reference 0ABV: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03M5: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABY: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AB1: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 073K: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AYG: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AYF: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABP: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AD4: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AYH: focused-seeds/geometry/S9-spaces-geom-conn-fibres-towards-normal/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 0A1C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8197–8197. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is proper,*
(2) *Y is integral (Spaces over Fields, Definition 0AD4) with generic point ξ,*
(3) *Y is normal,*
(4) *X is reduced,*
(5) *every generic point of an irreducible component of $|X|$ maps to ξ,*
(6) *we have $H^0(X_\xi, \mathcal{O}) = \kappa(\xi)$.*

Then $f_*\mathcal{O}_X = \mathcal{O}_Y$ and f has geometrically connected fibres.

Proof. Apply Theorem 0A1C to get a factorization $X \to Y' \to Y$. It is enough to show that $Y' = Y$. It suffices to show that $Y' \times_Y V \to V$ is an isomorphism, where $V \to Y$ is an étale morphism and V an affine integral scheme, see Spaces over Fields, Lemma 0AYH. The formation of Y' commutes with étale base change, see Morphisms of Spaces, Lemma 0ABP. The generic points of $X \times_Y V$ lie over the generic points of X (Decent Spaces, Lemma 0ABV) hence map to the generic point of V by assumption (5). Moreover, condition (6) is preserved under the base change by $V \to Y$, for example by flat base change (Cohomology of Spaces, Lemma 073K). Thus it suffices to prove the lemma in case Y is a normal integral affine scheme.

Assume Y is a normal integral affine scheme. We will show $Y' \to Y$ is an isomorphism by an application of Morphisms, Lemma 0AB1. Namely, Y' is reduced because X is reduced (Morphisms of Spaces, Lemma 0AYF). The morphism $Y' \to Y$ is integral by the theorem cited above. Since Y is decent and $X \to Y$ is separated, we see that X is decent too; to see this use Decent Spaces, Lemmas 03M5 and 0ABY. By assumption (5), Morphisms of Spaces, Lemma 0AYG, and Decent Spaces, Lemma 0ABV we see that every generic point of an irreducible component of $|Y'|$ maps to ξ. On the other hand, since Y' is the relative spectrum of $f_*\mathcal{O}_X$ we see that the scheme theoretic fibre Y'_ξ is the spectrum of $H^0(X_\xi, \mathcal{O})$ which is equal to $\kappa(\xi)$ by assumption. Hence Y' is an integral scheme with function field equal to the function field of Y. This finishes the proof. □